

Limits of Static Cubature for Deep ReLU Gaussian Expectations

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Abstract

Gaussian expectations of neural networks appear in uncertainty propagation, model analysis, and deterministic inference. When each network evaluation is expensive, a natural approach is to replace Monte Carlo samples with a fixed weighted rule. This paper studies the best possible performance of such rules for the infinite-width kernel of a dimension-256, depth-32 bias-free ReLU network.

Positive homogeneity reduces the Gaussian expectation to integration on the unit sphere. A complete system of 129 real mutually unbiased bases, generated by Kerdock structure and evaluated antipodally, gives a rule with 66,048 nodes. We prove that this rule is at most 0.0233242% above the optimum among all nonnegative mass-one rules with the same node budget. We also prove that arbitrary signed mass-one rules can reduce its kernel risk by at most 6.2940%. The first bound follows from an all-degree Delsarte certificate with a unique degree-five Hermite optimizer. The second uses harmonic moment constraints, matrix inertia, and an exact-rational witness. Together, the results show that fixed network-independent linear cubature is nearly exhausted for this kernel and budget.

1 Introduction

Many machine-learning calculations require an expectation of the form

$$\mu(f) = \mathbb{E}_{X \sim \mathcal{N}(0, I_d)}[f(X)]$$

The function f may be a deep neural network, so each evaluation can be expensive. Monte Carlo is general, but its convergence can be slow. Deterministic cubature instead chooses a fixed set of inputs and returns a weighted average of the outputs.

This paper studies a basic limit of that strategy. The nodes and weights are fixed before the network is realized. The estimator is linear in the observed outputs. The total number of network evaluations is fixed. Within this class, how much can any rule improve on a strong structured design?

The design considered here comes from real mutually unbiased bases. In dimension 256, a complete system contains 129 orthonormal bases. Evaluating every basis vector and its antipode gives 66,048 points on the sphere. The construction has a small algebraic set of pairwise inner products. This makes its error both low and exactly analyzable.

Our main result is a sharp class-level bound. Among all nonnegative mass-one rules with at most 66,048 nodes, the complete Kerdock rule is at most 0.0233242% above the optimum. Allowing signed weights enlarges the class, but the largest possible reduction in Kerdock risk is still at most 6.2940%.

The contribution is not a new search heuristic. It is a boundary on an entire estimator family. The proof connects neural-network Gaussian-process kernels, spherical harmonic energy, Delsarte linear programming, and inertia bounds for signed atomic measures.

The paper has three parts.

1. We express ensemble integration error as a spherical kernel discrepancy.
2. We prove near-optimality for nonnegative rules and a constant-factor bound for signed rules.
3. We give replayable exact and interval certificates for both results.

2 Problem setting

2.1 From Gaussian input to spherical integration

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be positively homogeneous of degree one:

$$f(rx) = rf(x), \quad r \geq 0$$

Write a standard Gaussian vector as $X = RU$, where U is uniform on $\mathbb{S}^{d-1}$ and $R = \|X\|_2$ is independent of U . Then

$$\mathbb{E}[f(X)] = \mathbb{E}[R] \int_{\mathbb{S}^{d-1}} f(u) d\sigma(u)$$

Bias-free ReLU networks satisfy this homogeneity exactly. The radial factor is known, so the numerical problem reduces to integration on the sphere.

2.2 The deep ReLU kernel

The centered infinite-width network output is a rotationally invariant Gaussian process. Its covariance on the sphere has the form

$$\mathbb{E}[F(u)F(v)] = K_{32}(u^\top v)$$

where

$$K_0(t) = t, \quad K_{\ell+1}(t) = \kappa(K_\ell(t))$$

and

$$\kappa(t) = \frac{\sqrt{1-t^2} + (\pi - \arccos t)t}{\pi}$$

The kernel has a normalized Gegenbauer expansion

$$K_{32}(t) = \sum_{\ell \geq 0} a_\ell G_\ell(t), \quad a_\ell \geq 0, \quad G_\ell(1) = 1$$

Each coefficient a_ℓ measures the energy in spherical harmonic degree ℓ .

2.3 Static linear cubature

A rule with at most N consolidated nodes is

$$Q(f) = \sum_{i=1}^n w_i f(u_i), \quad n \leq N, \quad u_i \in \mathbb{S}^{255}, \quad \sum_{i=1}^n w_i = 1$$

The nodes and weights are independent of the realized network and of all observed function values. We study two classes:

$$\mathcal{C}_+ = \{Q : w_i \geq 0\}, \quad \mathcal{C}_\pm = \{Q : w_i \in \mathbb{R}\}$$

Define the ensemble mean-squared integration error

$$\mathcal{R}(Q) = \mathbb{E} \left[\left(Q(F) - \int F d\sigma \right)^2 \right]$$

Because the rule is fixed before the Gaussian field is drawn,

$$\mathcal{R}(Q) = \iint K_{32}(u^\top v) d(\sigma - Q)(u) d(\sigma - Q)(v)$$

Thus the neural-network integration problem is exactly a kernel discrepancy problem on the sphere.

2.4 The complete Kerdock rule

A complete real mutually unbiased system in dimension $d = 256$ contains

$$B = d/2 + 1 = 129$$

orthonormal bases. The system consists of 128 Kerdock or Walsh-Hadamard chirp bases and the coordinate basis. Vectors from different bases have inner-product magnitude $1/16$.

The complete Kerdock rule Q_K uses every basis vector and its antipode. It gives equal weight to $N = 2Bd = 66,048$ nodes. Among ordered pairs, N have inner product 1, N have inner product -1 , and $4Bd(d-1)$ have inner product 0. The two values $1/16$ and $-1/16$ each occur $2B(B-1)d^2$ times.

This spectrum reduces $\mathcal{R}(Q_K)$ to five kernel evaluations and the constant Gegenbauer coefficient.

2.5 Related work

Delsarte linear programming and spherical designs provide the framework for the nonnegative bound [Delsarte et al., 1977, Schoenberg, 1942, Cohn and Kumar, 2007, Wendland, 2005]. Recent work develops general energy bounds for weighted spherical codes and designs [Borodachov et al., 2024]. Here the potential and budget are fixed: the depth-32 ReLU kernel and 66,048 nodes.

Kerdock codes, orthogonal spreads, and real mutually unbiased bases are closely related to extremal line sets and association schemes [Calderbank et al., 1997, Boykin et al., 2005, Abdukhalikov et al., 2009, Can et al., 2020]. Exact semidefinite bounds also establish packing optimality for the same family of antipodal Kerdock arrangements [Cohn et al., 2024]. Packing and kernel energy are different objectives.

Infinite-width neural networks induce compositional Gaussian-process kernels [Cho and Saul, 2009, Daniely et al., 2016, Lee et al., 2018, Matthews et al., 2018]. Kernel quadrature usually studies rates, node selection, or posterior uncertainty [Briol et al., 2019, Kanagawa et al., 2018]. This paper gives a finite-budget bound for a fixed deep neural kernel.

3 Main results

Let P_N be the infimum of $\mathcal{R}(Q)$ over $\mathcal{C}_+$ with at most N consolidated nonzero-weight nodes. Let S_N be the corresponding infimum over $\mathcal{C}_\pm$.

Theorem 1 (Nonnegative near-optimality). *For the dimension-256, depth-32 kernel and $N = 66,048$,*

$$1 \leq \frac{\mathcal{R}(Q_K)}{P_N} \leq 1.0002332417295004$$

This ratio means that Kerdock risk is at most 0.023324172950039% above the optimum.

Theorem 2 (Signed bound). *For the same kernel and node budget,*

$$S_N \geq 0.9370601683665084 \mathcal{R}(Q_K)$$

Equivalently,

$$\frac{\mathcal{R}(Q_K)}{S_N} \leq 1.0671673322143325$$

Signed rules can therefore reduce Kerdock risk by at most 6.29398316334916%.

Theorem 1 nearly determines the optimum over probability rules. Theorem 2 gives a broader bound that permits cancellation through negative weights.

Proposition 3 (Negative support count). *Consolidate duplicate nodes and remove zero weights. If a signed rule has at least 1,072 negative-weight support entries, its certified Kerdock-to-rule risk factor is below 1.05. If it has at least 4,160 negative-weight support entries, its certified lower bound is worse than Kerdock.*

The proposition concerns the number of negative support entries. It does not concern the total negative mass.

4 The nonnegative bound

The proof of Theorem 1 is a Delsarte lower bound that is optimized over every harmonic degree.

4.1 Auxiliary minorants

Let

$$h(t) = \sum_{\ell=0}^L c_\ell G_\ell(t)$$

be a polynomial such that

$$h(t) \leq K_{32}(t) \quad (-1 \leq t \leq 1), \quad c_\ell \geq 0 \quad (\ell \geq 1)$$

For every nonnegative mass-one rule,

$$\sum_{i,j} w_i w_j G_\ell(u_i^\top u_j) \geq 0 \quad (\ell \geq 1)$$

Also,

$$\sum_i w_i^2 \geq \frac{1}{N}$$

Hence

$$E_K(Q) \geq c_0 + \frac{K_{32}(1) - h(1)}{N}$$

where

$$E_K(Q) = \sum_{i,j} w_i w_j K_{32}(u_i^\top u_j)$$

Subtracting the constant kernel mean gives a lower bound on $\mathcal{R}(Q)$.

The certificate maximizes

$$\Phi(h) = c_0 + \frac{K_{32}(1) - h(1)}{N}$$

over all feasible Gegenbauer expansions.

4.2 The optimal minorant

Let t_1, t_2, t_3 be the roots of

$$22102t^3 + 21930t^2 - 87t - 85 = 0$$

Let $h_\star$ be the degree-five Hermite interpolant defined by

$$h_\star(t_j) = K_{32}(t_j), \quad h'_\star(t_j) = K'_{32}(t_j), \quad j = 1, 2, 3$$

Theorem 4 (All-degree auxiliary optimum). *The polynomial $h_\star$ is feasible. Its five nonconstant normalized Gegenbauer coefficients are strictly positive. It is the unique optimizer of Φ over finite feasible expansions and over the stated absolutely convergent nonnegative expansions.*

The proof has four steps.

4.2.1 Exact dual quadrature

The objective Φ defines a linear functional with rational moments through degree five. Its monic degree-three orthogonal polynomial is the cubic above. Exact root intervals and Lagrange formulas give a positive three-node quadrature rule that is exact through degree five.

4.2.2 Negative reduced costs above degree five

Exact integer recurrence checks degrees 6 through 14,658. A normalized Gegenbauer tail bound covers every larger degree. All omitted degrees have strictly negative reduced cost. Higher harmonic terms cannot improve the auxiliary objective.

4.2.3 Global minorant

A directed derivative certificate proves

$$K_{32}^{(6)}(t) > 0 \quad (-1 < t < 1)$$

The Hermite remainder formula gives

$$K_{32}(t) - h_\star(t) = \frac{K_{32}^{(6)}(\xi_t)}{6!} \prod_{j=1}^3 (t - t_j)^2 \geq 0$$

Separate directed certificates verify the two endpoints.

4.2.4 Primal-dual equality

A Krawczyk interval solve places all five nonconstant Gegenbauer coefficients above zero. Contact at the three quadrature nodes makes the primal and dual objectives equal. Strict reduced costs and the six interpolation conditions give uniqueness.

The resulting directed bound is

$$\mathcal{R}(Q) \geq 2.4330928587565938 \times 10^{-7} \quad (Q \in \mathcal{C}_+)$$

Comparison with the certified Kerdock-risk interval proves Theorem 1.

5 The signed bound

Negative weights can cancel harmonic moments. The signed proof therefore uses the inertia of harmonic moment matrices.

5.1 A positive-index inequality

For one harmonic comparison profile, let E be its node-evaluation matrix and let

$$W = \text{diag}(w_i)$$

The moment matrix is

$$M = E^T W E$$

Let the target matrix have trace $T > 0$ and squared Frobenius norm S_2 . Under the fixed block-trace constraints,

$$\|A - M\|_F^2 = \|M\|_F^2 - S_2$$

Lemma 5 (Positive-index bound). *If a symmetric matrix M has trace $T > 0$ and at most p positive eigenvalues, then*

$$\|M\|_F^2 \geq \frac{T^2}{p}$$

Proof. Let $\lambda_1^+, \dots, \lambda_p^+$ be the positive eigenvalues. Their sum is at least T . Cauchy-Schwarz gives

$$\|M\|_F^2 \geq \sum_i (\lambda_i^+)^2 \geq \frac{(\sum_i \lambda_i^+)^2}{p} \geq \frac{T^2}{p}. \quad \square$$

Congruence cannot increase the positive inertia of W . If a rule has q negative support entries, then M has at most $N - q$ positive eigenvalues. Therefore

$$\|A - M\|_F^2 \geq \frac{T^2}{N - q} - S_2$$

5.2 The harmonic witness

The certificate uses 146 nonnegative comparison profiles and kernel coefficients through degree 320. The coefficient allocation is rational and fixed. Each abstract rank floor is replaced by the positive-index floor above. The final sum is exact rational arithmetic once the directed lower endpoints of the kernel coefficients are supplied.

The certificate gives

$$\mathcal{R}(Q) \geq 2.2804861843861462 \times 10^{-7} \quad (Q \in \mathcal{C}_\pm)$$

This proves Theorem 2. Replacing $N - 1$ by $N - q$ gives Proposition 3.

6 Structure of the remaining signed gap

The signed bound leaves a 6.2940% interval. The current relaxation is sharp for abstract moment matrices. A single positive-semidefinite rank- N Gram matrix can attain every profile floor in the finite witness. Such a matrix need not come from evaluations at points on the sphere.

Atomic equality is more restrictive. It would force equal positive weights and a common zero condition across active comparison kernels. Two active profiles use consecutive Gegenbauer degrees 3 and 4 with different radial ratios. Consecutive Gegenbauer polynomials have no common zero. The older abstract floor is therefore not attained by an atomic point rule.

This identifies the source of the remaining gap. A tighter bound must encode point realizability beyond second-order moment matrices.

7 Implications

The nonnegative theorem leaves only about two parts in ten thousand between Kerdock and the optimum. At this kernel and node budget, further optimization of fixed nodes and nonnegative weights cannot produce a material improvement.

Signed weights create more room, but the certified maximum gain is still 6.2940%. Large negative supports make the bound worse. Any signed construction near the boundary must therefore use a small and carefully structured negative support.

The broader conclusion is simple. Static geometry already captures almost all available performance in this setting. A materially better estimator must use information that is not present in a fixed network-independent linear rule.

8 Verification

The artifact release contains the Kerdock pair-spectrum calculation, the all-degree Delsarte certificate, the Hermite and endpoint checks, the signed rational witness, the positive-index strengthening, and the negative-support hierarchy.

The core replay runs with

```
python scripts/run_core_verification.py
python scripts/check_release_strict.py
```

Directed intervals certify the kernel inequalities and coefficient bounds. Exact integer and rational arithmetic then aggregates the final bounds.

The release also includes the research ledger and the full chain of supporting audits. These records are separate from the theorem evidence.

9 Conclusion

We studied the best performance of static, network-independent linear cubature for a deep ReLU Gaussian-process kernel. The complete Kerdock rule uses 66,048 nodes. Its risk is at most 0.0233242% above the optimum among nonnegative mass-one rules. Even arbitrary signed mass-one rules can improve its risk by at most 6.2940% under the certified bound.

The result turns a search problem into a limit theorem. For this kernel and budget, the main gains available from fixed cubature geometry have already been captured.

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